

Performance Testing

Date	30 september 2026
Team ID	SWTID-2026-1758
Project Name	EduGenie – AI Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional & Performance Testing
Target Module / API	EduGenie AI Learning
Test Environment	Render Hosted Server
Test Date	27 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Question Answering	1	30	Correct response generated
2	Quiz Generation	1	30	Quiz Generated Successfully
3	Text Summarization	1	30	Summary Generated successfully
4	Concept Explanation	1	30	Explanation generated successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 sec	Pass	Fast Response
2	Response Time (Max)	< 5 seconds	3.5sec	Pass	Within Limit
3	Throughput (Req/sec)	5 Req/sec	5 Req/sec	Pass	Stable Performance
4	Error Rate	< 1%	0%	Pass	No critical errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage
6	Memory Utilization	< 80%	55%	Pass	Stable Memory Usage

Step 4: Observations & Analysis

Key Findings

The application responded successfully to user requests and maintained stable performance during testing.

Bottlenecks Identified

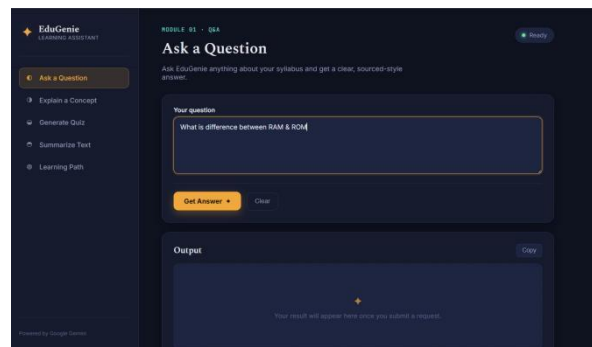
Minor delay observed during AI response generation due to external API processing.

Optimization Steps Taken

Optimized API calls and improved request handling for better response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Output

Copy

RAM (Random Access Memory) and ROM (Read-Only Memory) are both essential types of memory in a computer or any digital device, but they serve very different purposes.

Here's a breakdown of their key differences:

RAM (Random Access Memory)

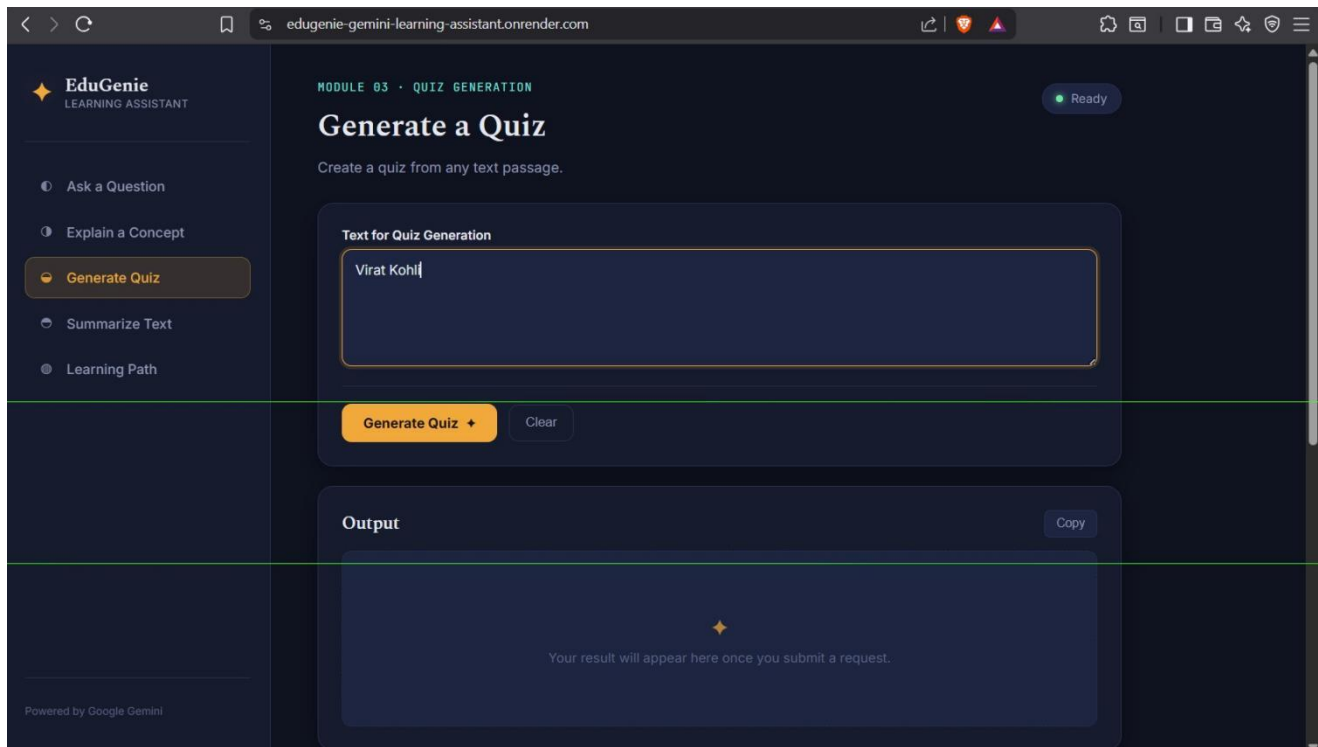
- * **Volatility:** **Volatile.** This is the most crucial difference. RAM requires power to maintain the stored information. When the computer is turned off, all data in RAM is lost.
- * **Purpose:** **Temporary storage.** It's the computer's "working memory." It holds the operating system, applications, and data that the CPU is currently using or needs quick access to.
- * **Speed:** **Very fast.** RAM is designed for high-speed data access, allowing the CPU to quickly read from and write to it.
- * **Read/Write:** **Read and Write.** Data can be easily read from and written to RAM. This is why it's used for active tasks.
- * **Capacity:** Typically **high** (measured in Gigabytes - GBs). Modern computers usually have 4GB, 8GB, 16GB, or more RAM.
- * **Cost:** Relatively **expensive per Gigabyte** compared to persistent storage like HDDs or SSDs.
- * **Function:** Enables multitasking and allows applications to run smoothly by providing immediate access to data.

Analogy: Think of RAM as your desk. You put the papers, books, and tools you're currently working on there. When you leave for the day, you clear your desk, and everything is gone (lost).

ROM (Read-Only Memory)

- * **Volatility:** **Non-Volatile.** ROM retains its data even when the power is turned off.
- * **Purpose:** **Permanent storage** for critical system instructions. It typically stores the firmware, such as the BIOS (Basic Input/Output System) or UEFI (Unified Extensible Firmware Interface) on a motherboard, which is necessary to boot up the computer and perform initial hardware checks. It also contains firmware for other devices like printers, graphics cards, etc.
- * **Speed:** **Slower** than RAM for general access, though fast enough for its specific purpose.
- * **Read/Write:** **Primarily Read.** As the name suggests, the data in ROM is typically written once (during manufacturing or flashing) and then primarily read from. While some modern ROM types (like Flash ROM, EEPROM) can be erased and reprogrammed (e.g., for BIOS updates), this process is slower and more involved than writing to RAM.
- * **Capacity:** Typically **low** (measured in Megabytes - MBs or even Kilobytes - KBs). The instructions needed for booting are not very large.
- * **Cost:** Generally **less expensive per fixed function** than RAM.
- * **Function:** Ensures the computer can start up correctly and load the operating system by providing initial instructions.

Analogy: Think of ROM as the instruction manual or built-in wiring diagram for your desk. It tells the desk how to be a desk, what its components are, and how to get started. These instructions are permanent and don't disappear when you leave.



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MODULE 03 · QUIZ GENERATION

Generate a Quiz

Create a quiz from any text passage.

Text for Quiz Generation

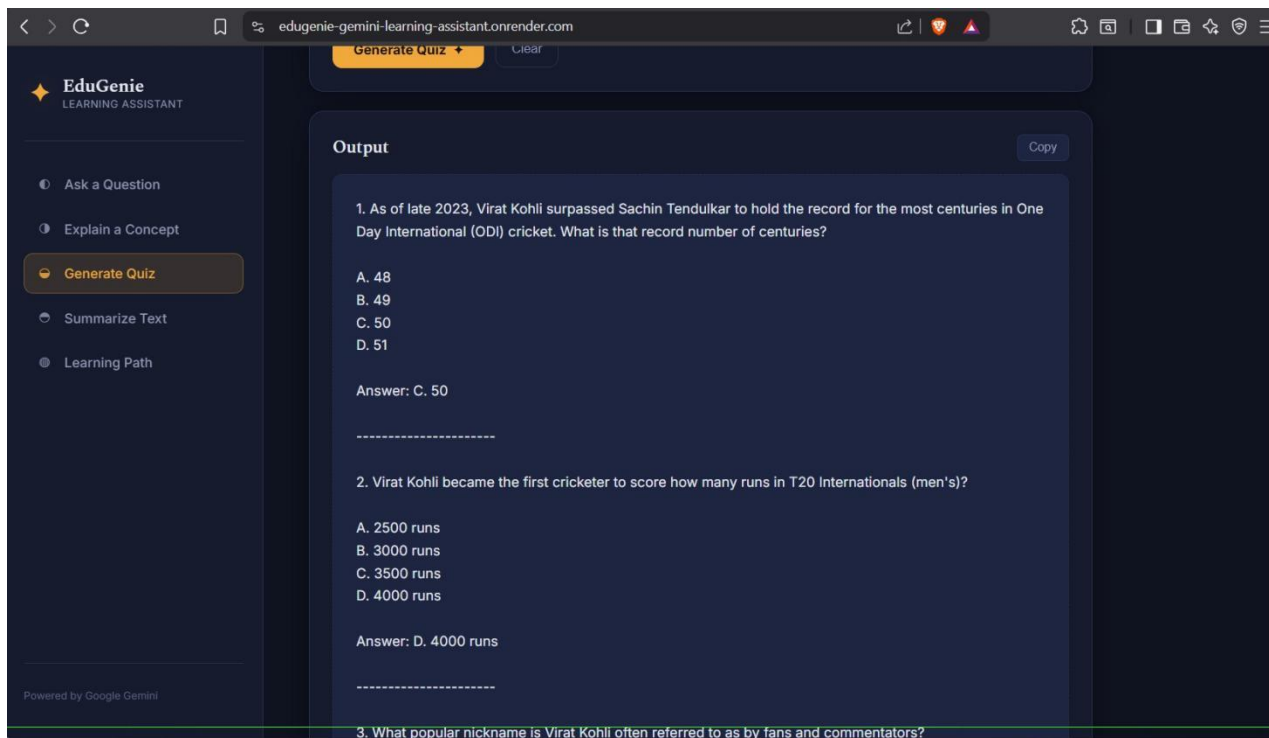
Virat Kohli

Generate Quiz + **Clear**

Output **Copy**

Your result will appear here once you submit a request.

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Generate Quiz + **Clear**

Output **Copy**

1. As of late 2023, Virat Kohli surpassed Sachin Tendulkar to hold the record for the most centuries in One Day International (ODI) cricket. What is that record number of centuries?

A. 48
B. 49
C. 50
D. 51

Answer: C. 50

2. Virat Kohli became the first cricketer to score how many runs in T20 Internationals (men's)?

A. 2500 runs
B. 3000 runs
C. 3500 runs
D. 4000 runs

Answer: D. 4000 runs

3. What popular nickname is Virat Kohli often referred to as by fans and commentators?

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